USN										

Roll No			

Shri Bhagawan Mahaveer Jain Educational and Cultural Trust's

JAIN COLLEGE OF ENGINEERING, BELAGAVI

(Approved by AICTE, New Delhi & Affiliated to VTU, Belagavi)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Academic Year 2026 - 2027

LABORATORY JOURNAL

Name :

Semester/Division :

Course Name :

Course Code :

LAB JOURNAL MARKS		LAB TEST MARKS		AVERAGE MARKS	
Maximum	Obtained	Maximum	Obtained	Maximum	Obtained

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JAIN COLLEGE OF ENGINEERING, BELAGAVI

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that Mr. / Ms. ________________________________ *bearing USN:* _______________ *has satisfactorily completed the list of experiments in the lab component for the course titled* **Internet of Things** *with Course Code:* ***BCS701*** *as partial fulfillment of the requirements for the Semester* ***7th*** *of the Bachelor of Engineering Degree Course prescribed by the Visvesvaraya Technological University, Belagavi during the academic year 2026- 2027.*

Faculty in Charge | **HOD,**
Dept. of CSE

In case of Lab External Exam

Examiner 1 ____________ *Examiner 2* ____________

INDEX

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

<u>Course Outcomes</u> (Course Skill Set):

At the end of the course the student will be able to:

CO1: Explain the evolution of IoT, IoT networking components, and addressing strategies in IoT.

CO2: Analyze various sensing devices and actuator types.

CO3: Demonstrate the processing in IoT.

CO4: Apply different connectivity technologies.

CO5: Elaborate the need for Data Analytics and Security in IoT.

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EXPT NO		EXPT.1	EXPT.2	EXPT.3	EXPT.4	EXPT.5	EXPT.6	EXPT.7	EXPT.8	EXPT.9	EXPT.10	EXPT.11	EXPT.12
DATE													
Attendance													
Journal Submission													
Execution													
Viva-Voce													
TOTAL MARKS													
STAFF SIGN													

AVERAGE MARKS	

JAIN COLLEGE OF ENGINEERING, BELAGAVI.

VISION

To be a university as a resource of solutions to diverse challenges of society by nurturing innovation, research & entrepreneurship through value-based education.

MISSION

- To provide work culture that facilitates effective teaching-learning process and lifelong learning skills
- To promote innovation, collaboration and leadership through best practices
- To foster industry-institute interaction resulting in entrepreneurship skills and employment opportunities

CORE VALUES

- Accountability
- Team Work
- Continuous Learning
- Holistic Development
- Competency
- Societal Responsibility

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VISION

To create a centre of excellence in teaching, learning and professional environment in the discipline of Computer Science and Engineering addressing emerging and existing challenges striving to meet the requirement of both industry and society.

MISSION

- To provide value-based quality technical education and training with continuous improvement in the discipline of Computer Science and Engineering.
- To develop professionals to meet the requirements of industry and society.
- To promote innovation, dedication, and hard work with ethically and professionally responsible behaviour.

PROGRAM SPECIFIC OBJECTIVES

Engineering Graduates will be able to:

- Identify the Problem, analyze, design, develop, test and implement solution using appropriate tools and technologies.
- Apply the knowledge to carry out innovative projects and transform ideas into working modules following the professional ethics and engineering principles.

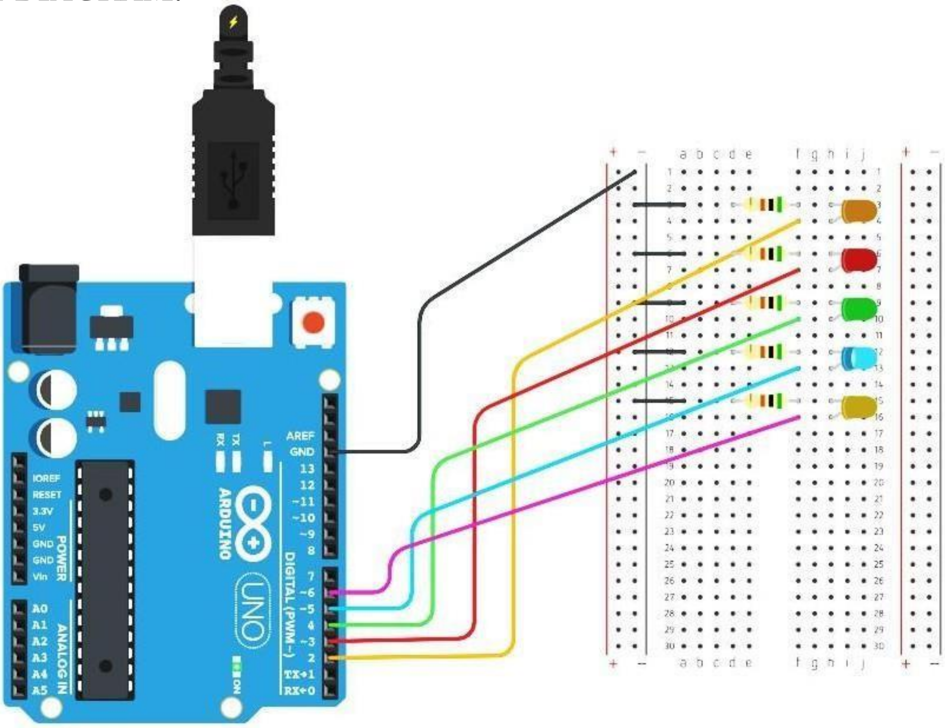
Exp. No. 1	Develop a program to blink 5 LED's back and forth.

AIM: Develop a program to blink 5 LEDs back and forth.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	6
3.	Bread Board	1
4.	LED	5
5.	Resistance (800 Ω)	5

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino

c) Open Arduino IDE then go to tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
void setup() {
    pinMode(2, OUTPUT);  // sets the digital pin 2-6 as output
    pinMode(3, OUTPUT);
    pinMode(4, OUTPUT);
    pinMode(5,  OUTPUT);
    pinMode(6, OUTPUT);
  }
void loop()  {
        digitalWrite(2, HIGH); // sets the digital pin 2 on
        delay(1000);           // waits for a second
        digitalWrite(2, LOW);  // sets the digital pin 2 off
        delay(100);
        digitalWrite(3, HIGH);
        delay(1000);
        digitalWrite(3, LOW);
        delay(100);
        digitalWrite(4, HIGH);
        delay(1000);
        digitalWrite(4, LOW);
        delay(100);
        digitalWrite(5, HIGH);
        delay(1000);
        digitalWrite(5, LOW);
        delay(100);
        digitalWrite(6, HIGH);
        delay(1000);
        digitalWrite(6, LOW); // Start Reverse Code
         delay(100);
        digitalWrite(5, HIGH);
        delay(1000);
        digitalWrite(5, LOW);
        delay(100);
        digitalWrite(4, HIGH);
        delay(1000);
        digitalWrite(4, LOW);
        delay(100);
        digitalWrite(3, HIGH);
        delay(1000);
        digitalWrite(3, LOW);
        delay(100);
        digitalWrite(2, HIGH);
        delay(1000);
}
```

Result: Successfully demonstrated blink 5 LEDs.

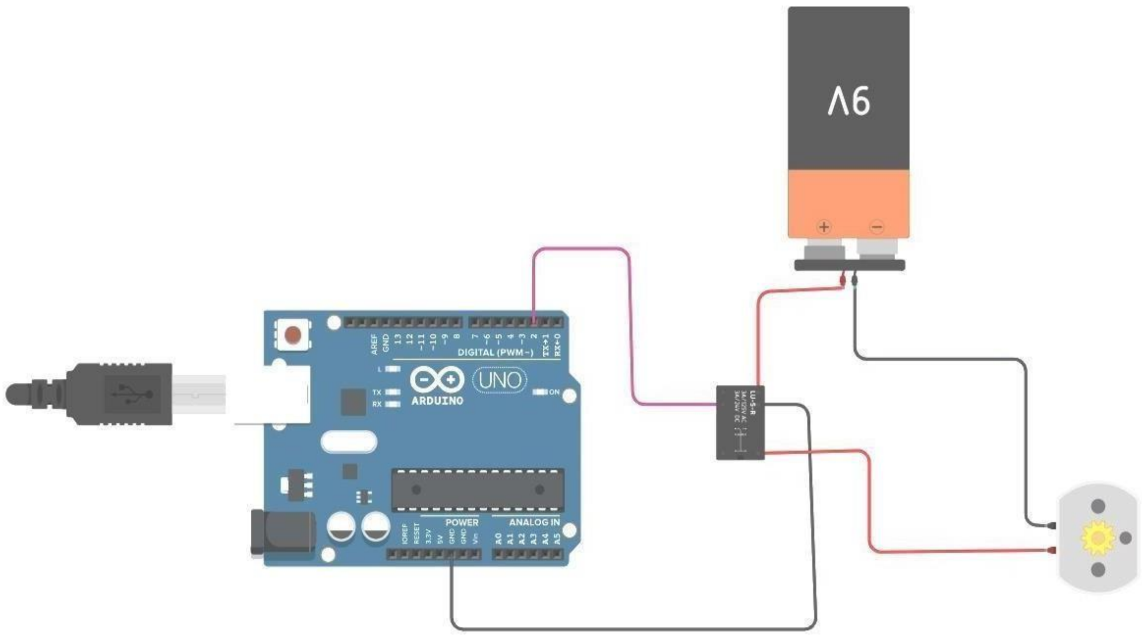
Exp. No. 2	Develop a program to interface a relay with Arduino board.

AIM: Develop a program to interface a relay with Arduino board.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	6
3.	Bread Board	1
4.	DC Motor	1
5.	Relay SPDT	1
6.	9V Battery	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino.

c) Open Arduino IDE then go to tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
void setup()
{
  pinMode(2, OUTPUT);
}

void loop()
{
  digitalWrite(2, HIGH);
  delay(10000); // Wait for 1000 millisecond(s)
  digitalWrite(2, LOW);
  delay(5000); // Wait for 1000 millisecond(s)
}
```

Result: Successfully demonstrated interface a relay with Arduino board

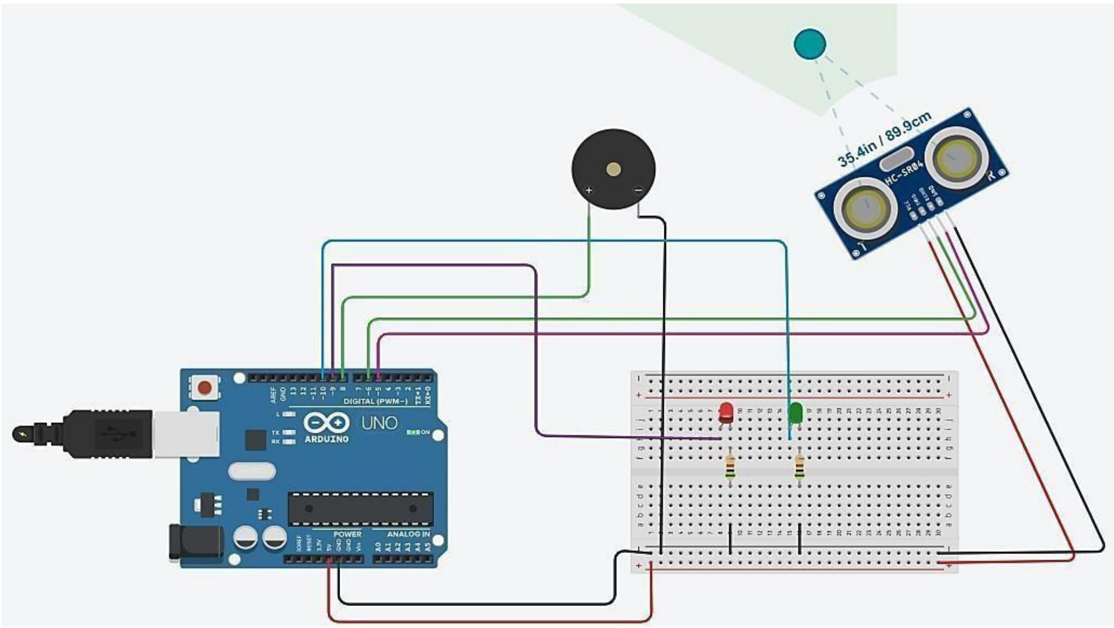
Exp. No. 3	Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

AIM: Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	6
3.	Bread Board	1
4.	LED	2
5.	Resistance (800 Ω)	2
6.	Ultrasonic Distance Sensor (4-pin)	1
7.	Piezo	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

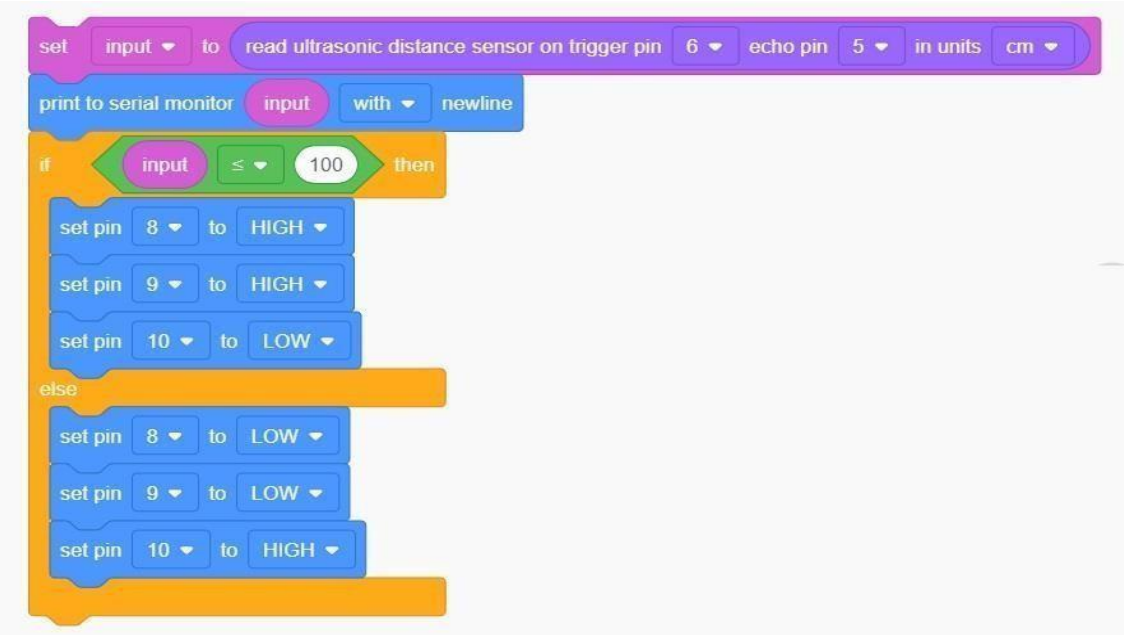
b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino.

c) Open Arduino IDE then go to tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:



Result: Successfully demonstrated deploy an intrusion detection system using Ultrasonic and sound sensors.

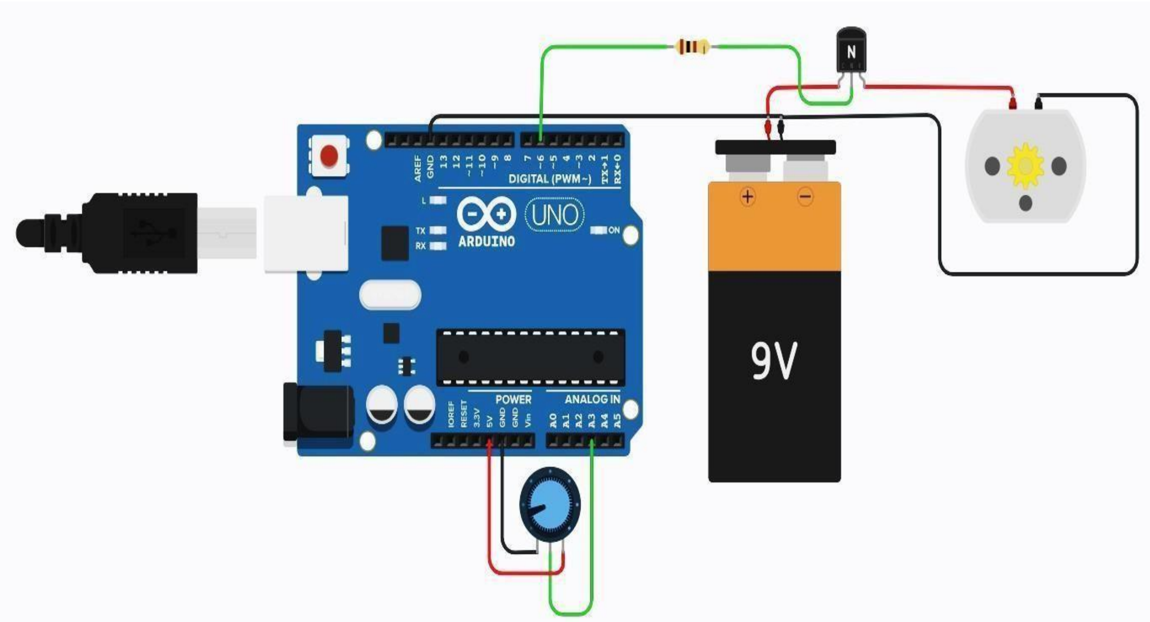
Exp. No. 4	Develop a program to control a DC motor with Arduino board.

AIM: Develop a program to control a DC motor with Arduino board.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	6
3.	Bread Board	1
4.	Resistance (800 Ω)	1
5.	DC Motor	1
6.	NPN Transistor (BJT)	1
7.	9V Battery	1
8.	250 kΩ Potentiometer	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
const int poten = A3;
 int var;

void setup()
{
pinMode(6, OUTPUT);
}

void loop()
{
var = analogRead(poten);
analogWrite(6,var);

}
```

Result: Successfully demonstrated control a DC motor with Arduino board

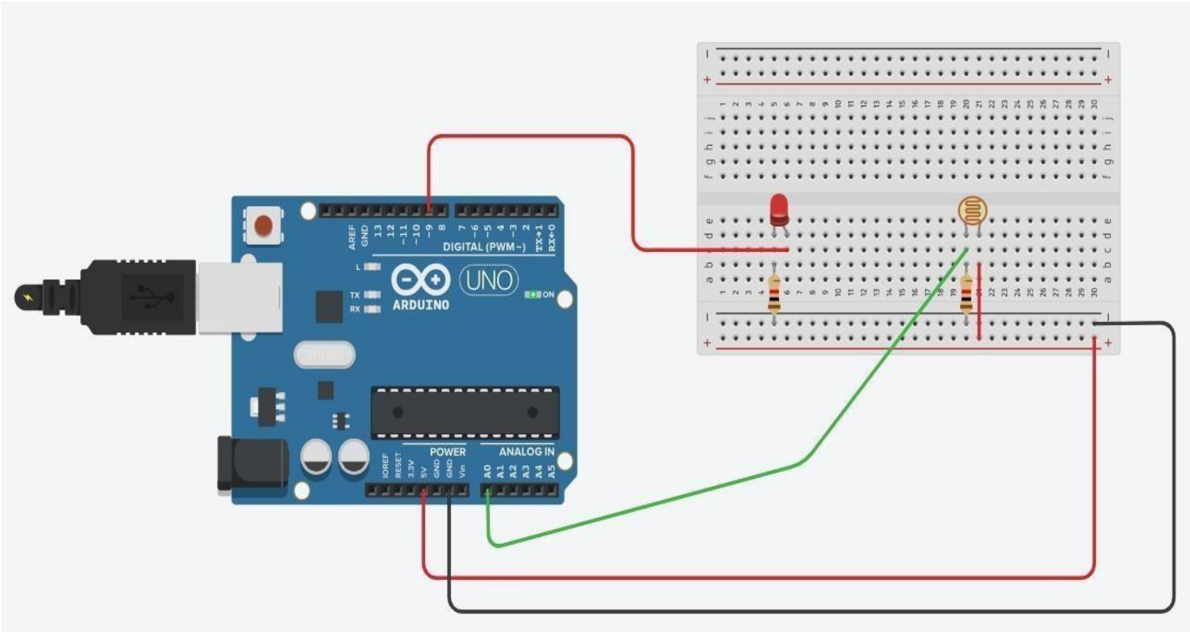
Exp. No. 5	Develop a program to deploy smart street light system using LDR sensor.

AIM: Develop a program to deploy smart street light system using LDR sensor.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	4
3.	Bread Board	1
4.	LED	1
5.	Resistance (850 Ω)	2
6.	LDR	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino.

c) Open Arduino IDE then go to tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
int sensorPin = A0;
int sensorValue = 0;
int led = 9;
void setup()
{
pinMode(led, OUTPUT);
Serial.begin(9600);
}
void loop()
{
sensorValue = analogRead(sensorPin);
Serial.println(sensorValue);
if(sensorValue < 100)
{
Serial.println("LED light on");
digitalWrite(led,HIGH);
delay(1000);
 }
digitalWrite(led,LOW);
delay(sensorValue);
}
```

Result: Successfully demonstrated deploy smart street light system using LDR sensor

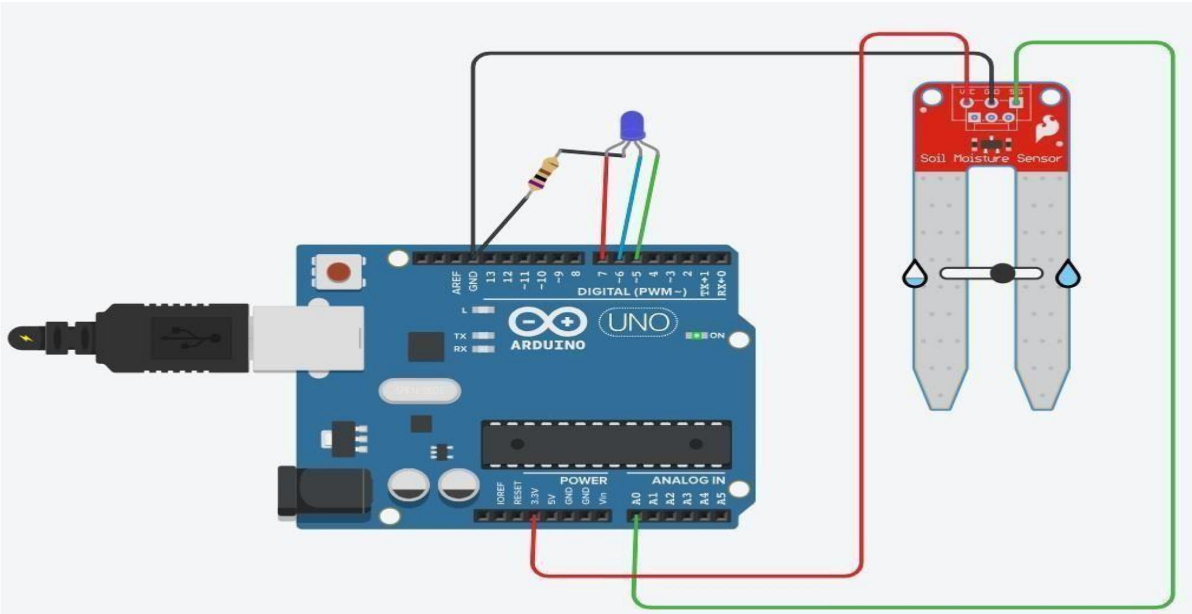
Exp. No. 6	Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

AIM: Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	8
3.	Bread Board	1
4.	LED	1
5.	Resistance (1KΩ)	1
6.	Moisture sensor (DHT22)	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino.

c) Open Arduino IDE then go to tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
int moistureValue;
float moisture_percentage;

void setup()
{
  pinMode(7, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(5, OUTPUT);
  Serial.begin(9600);
  }

void loop()
{
moistureValue = analogRead(A0);
  moisture_percentage = ((moistureValue/539.00)*100);
  if ( moisture_percentage>0 && moisture_percentage<25 )
  {
   digitalWrite(7,HIGH);
     digitalWrite(6,LOW);
     digitalWrite(5,LOW);
  }
  if ( moisture_percentage>25 && moisture_percentage<80 )
  {
  digitalWrite(7,LOW);
  digitalWrite(6,HIGH);
  digitalWrite(5,LOW);
  }
  if ( moisture_percentage>80 && moisture_percentage<100 )
  {
  digitalWrite(7,LOW);
  digitalWrite(6,LOW);
  digitalWrite(5,HIGH);
  }
  Serial.print("\nMoisture Value : ");
  Serial.print(moisture_percentage);
  Serial.print("%");
  delay(1000);
}
```

Result: Successfully demonstrated dry and wet waste with the Moisture sensor (DHT22)

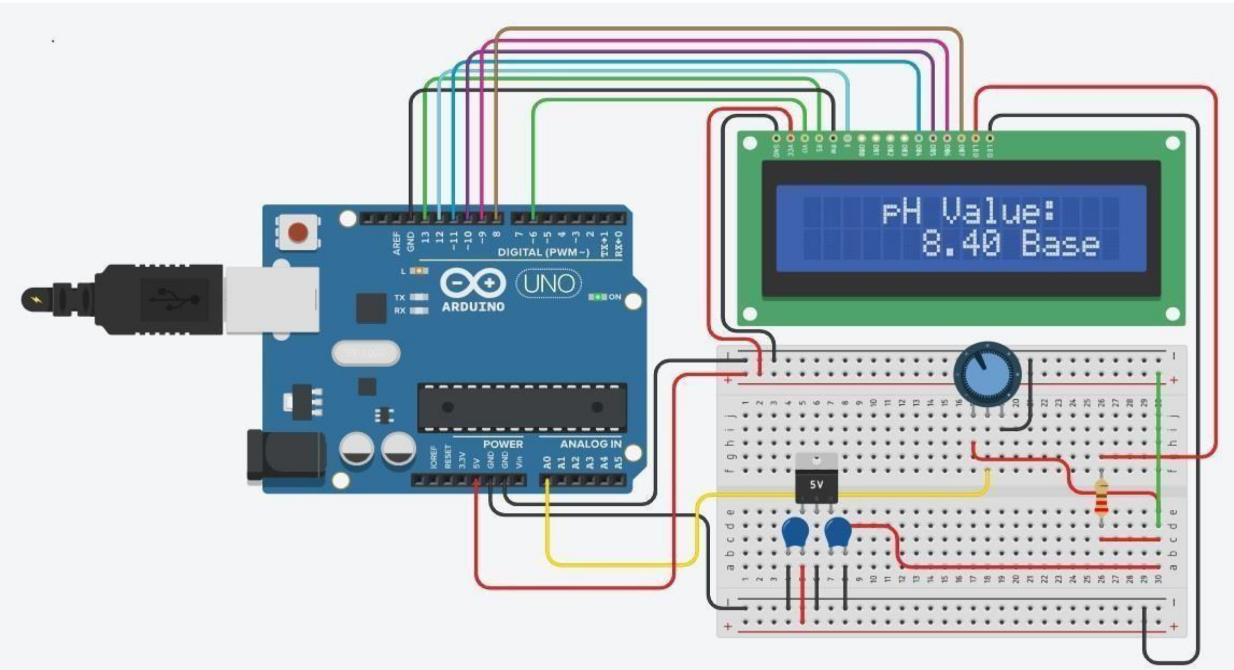
Exp. No. 7	Develop a program to read the pH value of a various substances like milk, lime and water.

AIM: Develop a program to read the pH value of a various substances like milk, lime and water.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	12
3.	Bread Board	1
4.	Resistance (220 Ω)	1
5.	LCD 16 x 2	1
6.	10 MΩ Potentiometer	1
7.	5V Regulator [LM7805]	1
8.	0.22 uF Capacitor	1
9.	0.1 uF Capacitor	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino.

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
#include<LiquidCrystal.h>
const int rs =13,en = 12,d4 =11,d5 =10,d6 =9,d7 =8;
LiquidCrystal lcd(rs,en, d4,d5,d6,d7);
int Contrast = 0;
void setup()
{
Serial.begin(9600);
analogWrite (6,Contrast);
lcd.begin(16,2);
lcd.setCursor(4,0);
lcd.print("pH Value:");
}

void loop()
{
int sensorValue = analogRead(A0);
float ph = sensorValue * (14.0/1023.0);
Serial.println(ph);
lcd.setCursor(6,1);
if (ph>0.0 && ph<5.0)
 {
 lcd.print (ph);
 lcd.print (" ACID");
 }
 if (ph>5.0 &&ph<7.0)
{
lcd.print (ph);
lcd.print (" Normal");
 }
if (ph>7.0 && ph<14.0)
  lcd.print (ph);
{
lcd.print (" Base");
}
}
```

Result: Successfully demonstrated read the pH value of a various substances like milk, lime and water

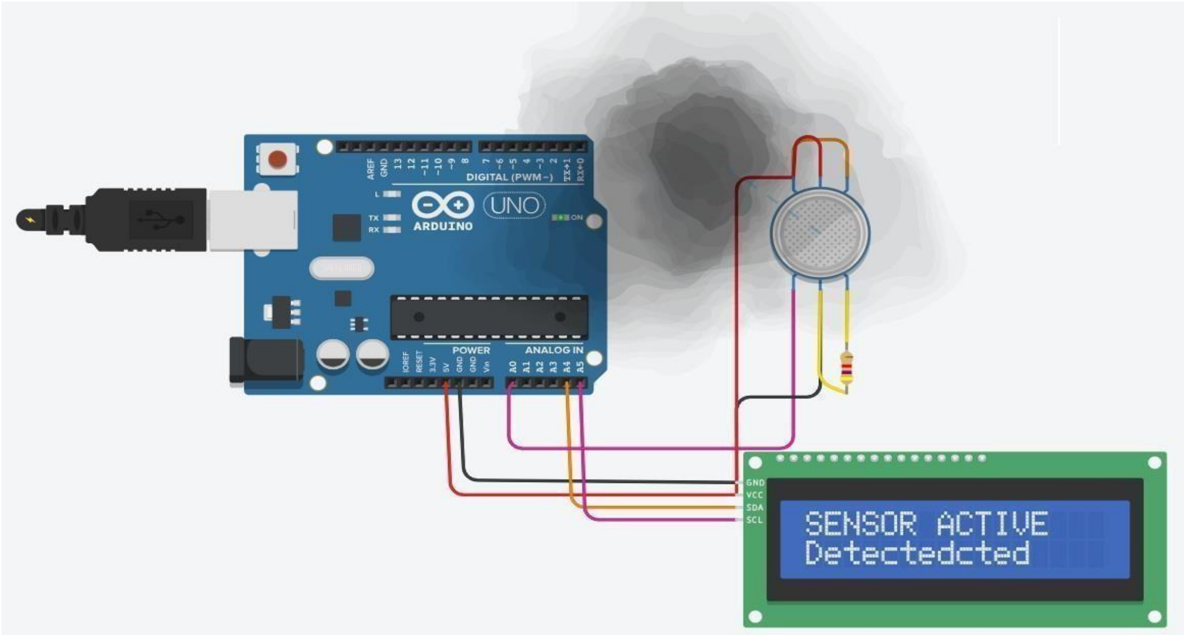
Exp. No. 8	Develop a program to detect the gas leakage in the surrounding environment.

AIM: Develop a program to detect the gas leakage in the surrounding environment.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	6
3.	Gas Sensor	1
4.	MCP23008-based, 32 (0x20) LCD 16 x 2 (I2C)	1
5.	Resistor (4.7 kΩ)	1
6.	Gas Sensor	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino.

c) Open Arduino IDE then go to tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

Result: Successfully demonstrated detect the gas leakage in the surrounding environment

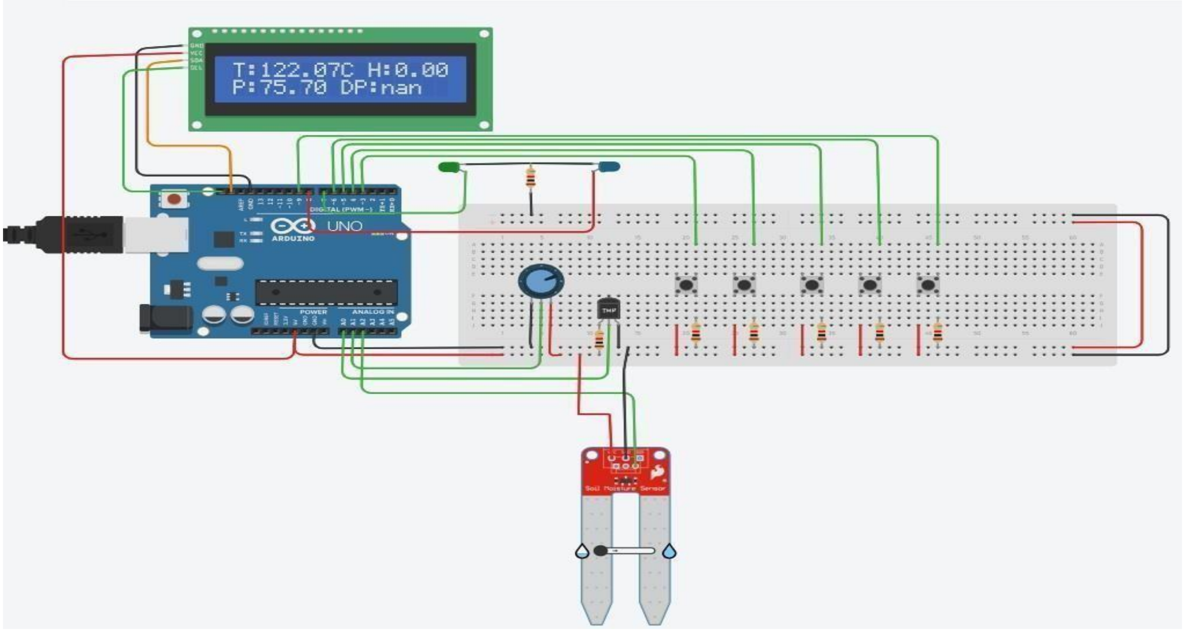
Exp. No. 9	Develop a program to demonstrate weather station readings using Arduino.

AIM: Develop a program to demonstrate weather station readings using Arduino.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	10
3.	Bread Board	1
4.	LED	2
5.	Resistance (1KΩ)	7
6.	LCD 16 x 2 (I2C)	1
7.	Potentiometer (250 kΩ)	1
8.	Temperature Sensor [TMP36]	1
9.	Soil Moisture Sensor	1
10.	Push Button	5

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
#include <Adafruit_LiquidCrystal.h>
#include <EEPROM.h>

Adafruit_LiquidCrystal lcd(0);
    const int tempPin = A0;
    const int humPin = A2;
    const int baroPin = A1;

    const int buttonPins[] = {3, 4, 5, 6,9};

    const int ledPins[] = {7, 8};

    float temperature, humidity, pressure, dewPoint;
    float minTemp = 100, maxTemp = -100;
    float minHumidity = 100, maxHumidity = 0; float
    minPressure = 1000, maxPressure = 0;

    float tempAlertThreshold = 30.0; float
    humAlertThreshold = 80.0;

    float tempHistory[10] = {0};
    float humHistory[10] = {0}; float
    pressHistory[10] = {0}; int
    historyIndex = 0;

    int currentScreen = 0;

    const int eepromStartAddr = 0; int
    eepromIndex = 0;

float calculateDewPoint(float temp, float hum) { float a
    = 17.27;
    float b = 237.7;
    float alpha = ((a * temp) / (b + temp)) + log(hum / 100.0); return (b *
    alpha) / (a - alpha);
    }


void logDataToEEPROM(float temp, float hum, float pres) {
   int addr = eepromStartAddr + eepromIndex * 12;
   if (addr + 12 > EEPROM.length()) eepromIndex = 0;
   EEPROM.put(addr, temp);
```

```
    EEPROM.put(addr + 4, hum);
    EEPROM.put(addr + 8, pres);
    eepromIndex++;
}

void readSensors()
{
    temperature = analogRead(tempPin) * 5.0 / 1024.0 * 100.0;
    humidity = map(analogRead(humPin), 0, 1023, 0, 100);
    pressure = analogRead(baroPin) / 10.0;
    dewPoint = calculateDewPoint(temperature, humidity);

    if (temperature < minTemp) minTemp = temperature;
    if (temperature > maxTemp) maxTemp = temperature;
    if (humidity < minHumidity) minHumidity = humidity;
    if (humidity > maxHumidity) maxHumidity = humidity;
    if (pressure < minPressure) minPressure = pressure;
    if (pressure > maxPressure) maxPressure = pressure;

    tempHistory[historyIndex] = temperature;
    humHistory[historyIndex] = humidity;
    pressHistory[historyIndex] = pressure;
    historyIndex = (historyIndex + 1) % 10;

    logDataToEEPROM(temperature, humidity, pressure);
}

void checkAlerts()
 {
    if (temperature > tempAlertThreshold)
    {
    digitalWrite(ledPins[0], HIGH);
    }
    else
    {
        digitalWrite(ledPins[0], LOW);
    }

    if (humidity > humAlertThreshold)
     {
    digitalWrite(ledPins[1], HIGH);
    }
    else
    {
        digitalWrite(ledPins[1], LOW);
    }
```

```
}

void exportData()
{
Serial.println("Exporting data:");
  for (int i = 0; i < eepromIndex; i++)
   {
  int addr = eepromStartAddr + i * 12;
    float temp, hum, pres;
    EEPROM.get(addr, temp);
    EEPROM.get(addr + 4, hum);
    EEPROM.get(addr + 8, pres);
    Serial.print("T:");
    Serial.print(temp);
    Serial.print(" H:");
    Serial.print(hum);
    Serial.print(" P:");
    Serial.println(pres);
  }
}

void displayGraph()
 {
lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("T:");
  for (int i = 0; i < 10; i++) {
    lcd.print(tempHistory[i] > tempAlertThreshold ? "*" : ".");
  }
  lcd.setCursor(0, 1);
  lcd.print("H:");
  for (int i = 0; i < 10; i++) {
    lcd.print(humHistory[i] > humAlertThreshold ? "*" : ".");
  }
}
String forecast = "N/A";

void calculateWeatherForecast()
{
  float pressureChange = pressHistory[9] - pressHistory[6];
  float humidity = humHistory[9];
  float temperature = tempHistory[9];

  if (pressureChange < -2.0 && humidity > 70)
   {
  forecast = "Rain expected";
```

```
    }
    else if (pressureChange > 2.0 && humidity < 50)
    {
    forecast = "Clear skies";
    }
     else if (temperature > 30.0)
    {
    forecast = "Hot weather";
    }
     else if (temperature < 5.0)
    {
     forecast = "Cold, frost";
    }
    else
    {
        forecast = "Stable";
    }
}
void displayForecast()
{
lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Forecast:");
    lcd.setCursor(0, 1);
    lcd.print(forecast);
}
void updateDisplay()
{
 lcd.clear();
    if (currentScreen == 0)
     {
    lcd.setCursor(0, 0);
        lcd.print("T:");
        lcd.print(temperature);
        lcd.print("C H:");
        lcd.print(humidity);
        lcd.setCursor(0, 1);
        lcd.print("P:");
        lcd.print(pressure);
        lcd.print(" DP:");
        lcd.print(dewPoint);
    }
    else if (currentScreen == 1)
    {
        lcd.setCursor(0, 0);
```

```
        lcd.print("Tmin:");
        lcd.print(minTemp);
        lcd.print(" Tmax:");
        lcd.print(maxTemp);
        lcd.setCursor(0, 1);
        lcd.print("Hmin:");
        lcd.print(minHumidity);
        lcd.print(" Hmax:");
        lcd.print(maxHumidity);
    }
    else if (currentScreen == 2)
    {
    displayGraph();
    }
    else if (currentScreen == 3)
    {
    exportData();
    lcd.setCursor(0, 0);
        lcd.print("Export Complete");
    }
   else if (currentScreen == 4)
   {
    displayForecast();
    }
}

void handleButtonPress()
{
for (int i = 0; i < 5; i++)
{
        if (digitalRead(buttonPins[i]) == HIGH)
         {
            currentScreen = i;
            updateDisplay();
            delay(300);
        }
    }
}

void setup()
{
lcd.begin(16, 2);
    lcd.setBacklight(1);
    Serial.begin(9600);
```

```
    for (int i = 0; i < 4; i++) pinMode(buttonPins[i], INPUT_PULLUP);
    for (int i = 0; i < 2; i++) pinMode(ledPins[i], OUTPUT);

    lcd.setCursor(0, 0);
    lcd.print("Weather Station");
    delay(2000);
    lcd.clear();
}

void loop()
{
 readSensors();
    checkAlerts();
    updateDisplay();
    handleButtonPress();
    calculateWeatherForecast();

    delay(2000);
}
```

Result: Successfully demonstrated weather station readings using Arduino

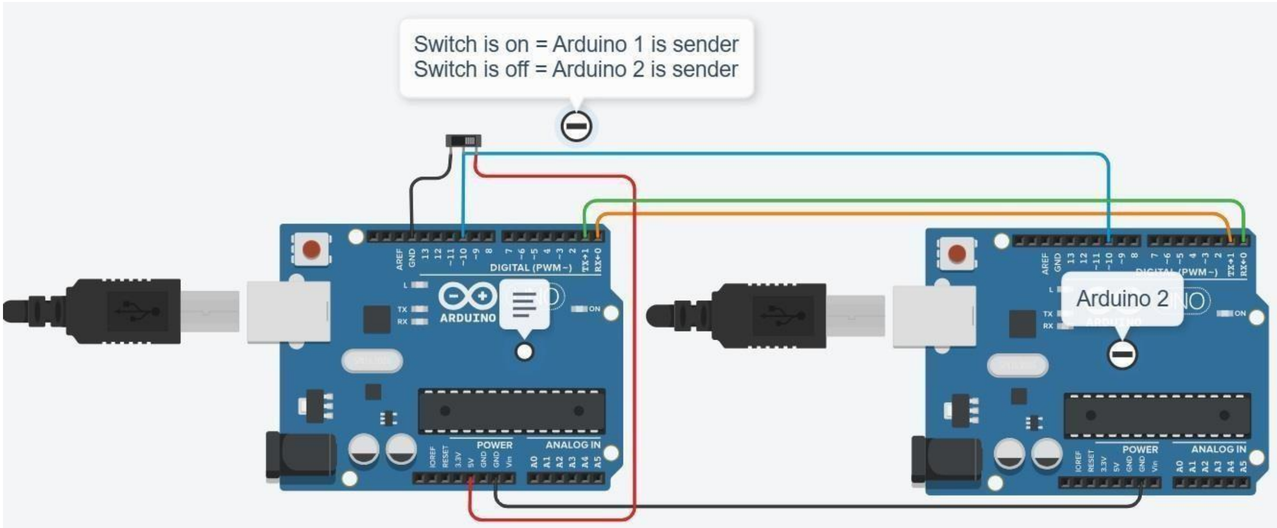
Exp. No. 10	Develop a program to setup a UART protocol and pass a string through the protocol.

AIM: Develop a program to setup a UART protocol and pass a string through the protocol.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	2
2.	Jumper Cable	7
3.	Slide switch	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

f) Then go to code serial monitor to check working output

CODE:

```
const int MAX_LEN =30;
char sendMsg[MAX_LEN] = "Hello i'm Arduino-2\n";
char receiveMsg[MAX_LEN];
int switchState = 1; // Active HIGH input
int switch_pin = 10;

void setup()

{

 Serial.begin(9600);
  pinMode(switch_pin, INPUT);
}

void loop()
 {
  switchState = digitalRead(switch_pin);

  if (switchState == HIGH)
   {
    // Receive if switch is off
    if (Serial.available() > 0)
    {
      int len = Serial.parseInt();        // Read the length as number
      Serial.read();                      // Consume the newline after the number

      int n = Serial.readBytes(receiveMsg, len);
      receiveMsg[n] = '\0';               // Null-terminate the received string

      Serial.print("Message = ");
      Serial.println(receiveMsg);
    }
  }
  else
  {
   // Send if switch is on
    int len = strlen(sendMsg);
    Serial.write(receiveMsg, len); // send raw message

    delay(1000);
  }
}
```

Result: Successfully demonstrated UART using two Arduino

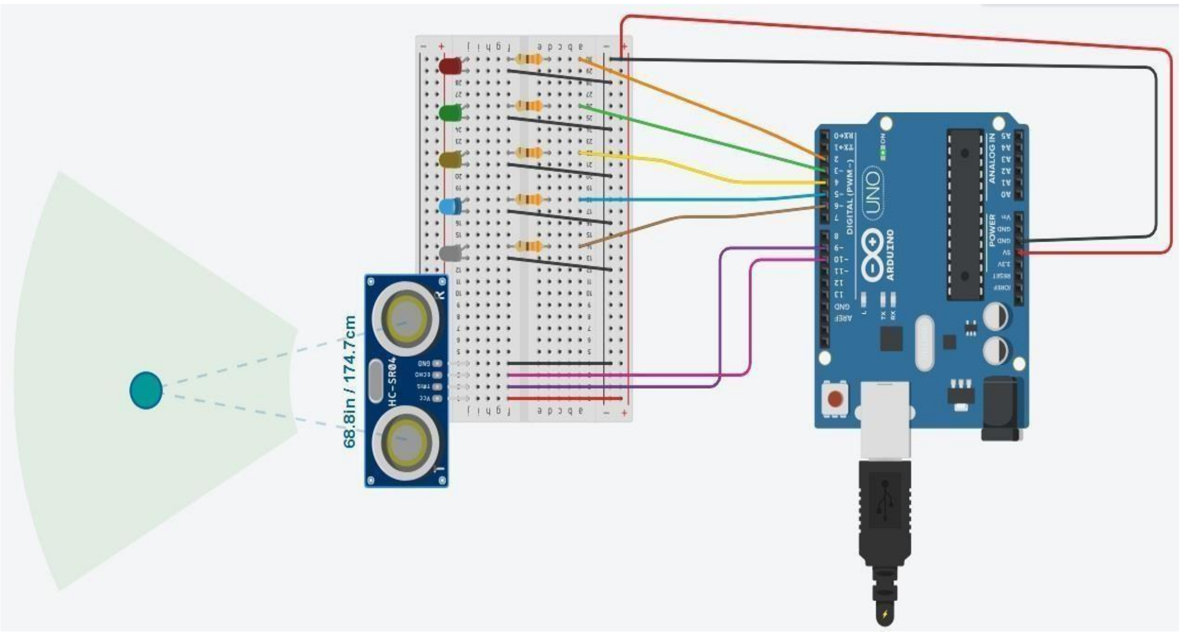
Exp. No. 11	Develop a water level depth detection system using Ultrasonic sensor.

AIM: Develop a water level depth detection system using Ultrasonic sensor.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	17
3.	Bread Board	1
4.	LED	5
5.	Resistance (800 Ω)	5
6.	Ultrasonic Distance Sensor (4-pin)	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
int trigPin=9;
int echoPin=10;
int a=2,b=3,c=5,d=6,e=4;
long dist;
long ival;
void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(a,OUTPUT);
  pinMode(b,OUTPUT);
  pinMode(e,OUTPUT);
  pinMode(c,OUTPUT);
  pinMode(d,OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  ival=pulseIn(echoPin,HIGH);
  dist=(ival/2)/29.154;
  Serial.print("dist:");
  Serial.print(dist);
  Serial.println("CM");

  if(dist<=50)
  {
   digitalWrite(a,HIGH);
  digitalWrite(b,LOW);
  digitalWrite(e,LOW);
  digitalWrite(c,LOW);
  digitalWrite(d,LOW);
  }
    else if(dist<=100)
    {
    digitalWrite(a,LOW);
    digitalWrite(b,HIGH);
```

```
digitalWrite(e,LOW);
digitalWrite(c,LOW);
digitalWrite(d,LOW);
}
else if(dist<=150)
{
digitalWrite(a,LOW);
digitalWrite(b,LOW);
digitalWrite(e,HIGH);
digitalWrite(c,LOW);
digitalWrite(d,LOW);
}
else if(dist<=200)
{
 digitalWrite(a,LOW);
digitalWrite(b,LOW);
digitalWrite(e,LOW);
digitalWrite(c,HIGH);
digitalWrite(d,LOW);
}
else if(dist<=250)
{
digitalWrite(a,LOW);
digitalWrite(b,LOW);
digitalWrite(e,LOW);
digitalWrite(c,LOW);
digitalWrite(d,HIGH);
  }
else
{
  digitalWrite(a,HIGH);
digitalWrite(b,HIGH);
digitalWrite(e,HIGH);
digitalWrite(c,HIGH);
digitalWrite(d,HIGH);
}
delay(50);
}
```

Result: Successfully demonstrated water level depth detection system using Ultrasonic sensor

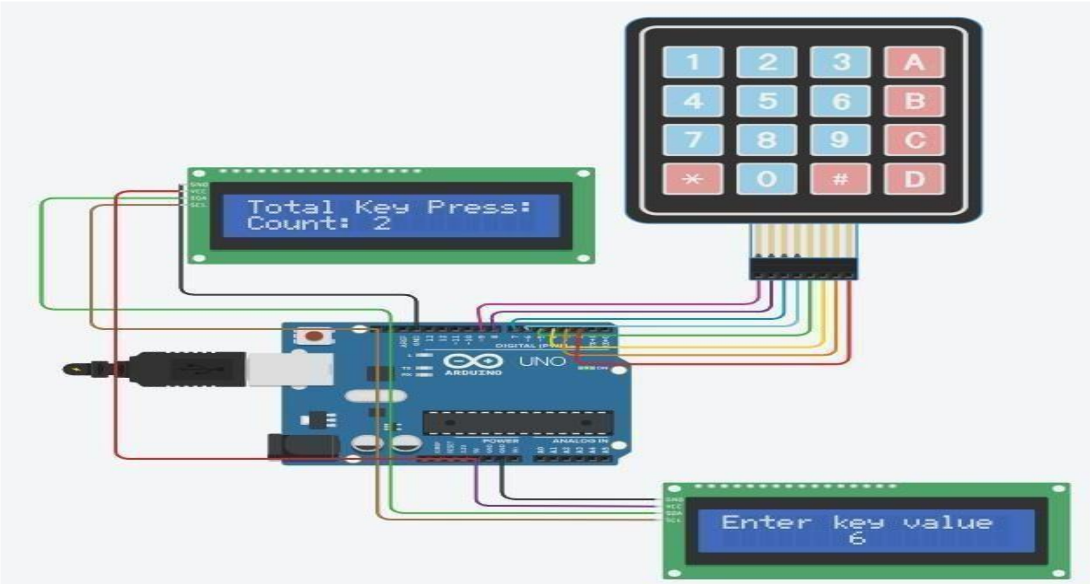
Exp. No. 12	Develop a program to simulate interfacing with the keypad module to record the keystrokes.

AIM: Develop a program to simulate interfacing with the keypad module to record the Keystrokes.

COMPONENT:

S.NO.	Name	Quantity
1.	Arduino Uno	1
2.	Jumper Cable	8
3.	Bread Board	1
4.	Keypad 4x4	1
5.	LCD 16 x 2 (I2C)	2

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
#include <Adafruit_LiquidCrystal.h>
#include <Keypad.h>

// LCD 1 – Shows key pressed
Adafruit_LiquidCrystal lcd_1(0x20);

// LCD 2 – Shows total key count
Adafruit_LiquidCrystal lcd_2(0x21);

const byte ROWS = 4;
const byte COLS = 4;

char hexaKeys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'},
};

byte rowPins[ROWS] = {9, 8, 7, 6};
byte colPins[COLS] = {5, 4, 3, 2};

Keypad customKeypad = Keypad(makeKeymap(hexaKeys), rowPins,
colPins, ROWS, COLS);

int keyPressCount = 0;

void setup() {
  Serial.begin(9600);

  // Initialize both LCDs
  lcd_1.begin(16, 2);
  lcd_1.setBacklight(1);
  lcd_1.print("Enter key value");
  lcd_1.setCursor(0, 1);

  lcd_2.begin(16, 2);
  lcd_2.setBacklight(1);
  lcd_2.print("Total Key Press:");
  lcd_2.setCursor(0, 1);
  lcd_2.print("Count: 0");
}
```

```
void loop()
 {
  char customKey = customKeypad.getKey();
  if (customKey) {
  Serial.println(customKey);

    // Display key on LCD 1
    lcd_1.setCursor(7, 1);
    lcd_1.print(" ");
    lcd_1.setCursor(7, 1);
    lcd_1.print(customKey);

    // Increment and show count on LCD 2
    keyPressCount++;
    lcd_2.setCursor(0, 1);
    lcd_2.print("Count: ");
    lcd_2.print(keyPressCount);
    lcd_2.print("   "); // Clear extra digits if count goes down

    delay(100);
   }
 }
```

Result: Successfully demonstrated simulate interfacing with the keypad module to record the keystrokes.